CHORUS-Skid SGH-1: A Proof-of-Concept Framework for Pressure-Retarded Osmosis on Anthropogenic Brine Gradients with Acoustic Harvest and Column-Scale Energy Accounting

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CHORUS Research — *differential-harness*

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Abstract

Coastal and inland desalination plants discharge hypersaline brine while treated effluent remains at near-fresh salinity. We present **CHORUS** (Columnar Harvest of Osmotic, Rhizospheric, Orographic, and Solar flux) as a multi-physics accounting framework for a 1 km^2 coastal parcel, and **CHORUS-Skid SGH-1** as a bench-scale pressure-retarded osmosis (PRO) harness targeting 10 W from brine (1400 mol/m^3) against wastewater (5 mol/m^3). Van't Hoff thermodynamics, Kim–Baker optimal pressure, and solution-diffusion transport yield $\Delta\pi = 6.92$ MPa, $\Delta P^* = 34.6$ bar, and $A_{mem} = 0.72$ m^2. Monte Carlo column integration ($N = 8000$) gives median 22.76 MW/km^2 (P10–P90: 19.6–26.5 MW), dominated by evaporatively coupled photovoltaics. Default L_p predicts 1.66 W versus the 10 W design target; inverse sizing gives $L_p^* \approx 6.0\times10^{\blacksquare 12}$ m/(Pa·s). We separate defensible near-term claims (PRO, acoustic harvest, ultrasonic CP assist) from exploratory Telluric Storm Coupling and atmospheric routing. Artifacts are open-source in differential-harness.

Keywords: pressure-retarded osmosis; salinity-gradient power; blue energy; concentration polarization; acoustic energy harvest; coastal energy systems

1. Introduction

1.1 Motivation

Reverse-osmosis desalination produces low-salinity permeate and reject brine often exceeding 7–8 wt% NaCl. Treated wastewater effluent remains at single-digit mol/m³ salinity. The Gibbs free energy of mixing between these streams is routinely dissipated rather than converted to work. Pressure-retarded osmosis (PRO) extracts hydraulic work as water permeates from feed into a pressurized draw compartment. Estuary RED demonstrations approach 15 W/m² electrical density; sidestream PRO on brine/effluent pairs offers higher $\Delta\pi$ at the cost of fouling and concentration polarization (CP).

1.2 Contributions

• Unified mathematical blueprint (Layers A–F) with explicit no-over-unity postulate.

• Executable SymPy/NumPy/SciPy proof exporting chorus_results.json.

• SGH-1 hardware: 12-plate PRO stack, OpenSCAD CAD, BOM, bench protocol T0/T1/T2.

• AEH-1 acoustic module: harvest (Mode A) and ultrasonic CP assist (Mode B).

• Honest power-density hierarchy: PV–hydro ■ blue energy ■ MEG/SMFC.

2. Theoretical framework

2.1 Osmotic driving force

For NaCl with van't Hoff factor $i = 2$: $\pi = iRTc$ and $\Delta\pi = \pi_{draw} - \pi_{feed}$. Nernst potential $E_N = (RT/F) \ln(c_{draw}/c_{feed})$. Estuary reference (600/20 mol/m³): $E_N = 87.39$ mV, $V_{stack} = 4.37$ V, $\Delta\pi = 2.876$ MPa. SGH-1 brine pair (1400/5): $\Delta\pi = 6.916$ MPa (~2.4× estuary).

2.2 PRO transport and Kim–Baker optimum

Water flux V■_w = $L_p A(\Delta\pi - \Delta P)$. Optimal hydraulic pressure $\Delta P^* = \Delta\pi/2$. Equivalent electrical power $P = \eta_{mem} \eta_{hyd} \rho$ V■_w ΔP. Operating band 0.4–0.6 $\Delta\pi$ for the bench skid.

2.3 Column balance (CHORUS)

Quantity	Value
P_column median	22.76 MW
P_column P10 / P90	19.63 / 26.46 MW
P_pv'' (evap-cooled)	187.9 W/m²
P_blue'' (estuary)	15.0 W/m²
P_MFC''	36.75 µW/m²

PV–hydro dominates the column median; osmotic interface power is a credible baseload supplement, not the land-area leader.

2.4 Concentration polarization and ultrasonic assist

Film model: $c_w/c_b = \exp(J_w/k_m)$. Effective driving force scales as $1/(c_w/c_b)_{outlet}$. AEH Mode B applies 28 kHz ultrasound to disrupt the polarization layer, modeled as multiplicative gain g on water permeability L_p. Net power $P_{net} = P_{PRO}(g) - P_{US} - P_0$. At $g = 1.35$ and $P_{US} = 1.5$ W/m², net benefit requires bench-validated flux gain exceeding parasitic driver load.

2.5 Telluric Storm Coupling (TSC)

Three-node conductance network (atmosphere, soil, estuary): $G\psi = I$. Dissipated power P_{TSC} = ψ■Gψ. TSC routes charge between high-impedance harvesters (MEG, SMFC); it does not create new source power. Illustrative conductances are order-of-magnitude placeholders pending geophysical instrumentation.

2.6 Rhizospheric and atmospheric bounds

Butler–Volmer kinetics bound microbial fuel-cell current; piezoelectrotrophy couples mechanical stress to bioelectrogenesis at µW/m² class (notebook: $P_{MFC}'' = 36.7$ µW/m²). Fair-weather global circuit $P_{glob} \approx 0.6$ GW implies areal mean ~10■■ W/m² — relevant for routing narratives, not land-area harvest.

3. CHORUS-Skid SGH-1 system

SGH-1 integrates a 12-plate PRO core (200×300 mm active area), AEH-1 piezo/ultrasonic panel, CHORUS enclosure for future moist-thermal ports, and DAQ for pressure, conductivity, flow, and voltage.

3.1 Sizing results

Parameter	Value
P_target	10.0 W
P''_design	8.0 W/m²
A_mem	0.72 m² (12 plates)
Δπ	6.916 MPa
ΔP*	34.58 bar
Q_feed (model)	0.149 L/min
Frame (L×W×H)	664×480.0×900 mm

3.2 Dimensionless groups

Group	Value
Π■ = ΔP/Δπ	0.500
Π■ = Pe	0.692
Π■ = P''A/P_target	0.576
Π■ = P_sim/P_target	0.166
L_p* for 10 W	6.03e-12 m/(Pa·s)
P_sim	1.66 W

4. Methods

Software: CHORUS_physics_proof.ipynb; simulation/pro_cycle.py, sizing.py, membrane_transport.py, ultrasonic_cp_gain.py; run_sizing.py; pi_groups.py. Hardware: 23 OpenSCAD parts, BUILD_BLUEPRINT.md, SGH1_TEST_PROTOCOL.md. Materials: SS316 wetted paths per MATERIALS_SPEC.md.

5. Discussion

PRO on anthropogenic brine leverages existing desal infrastructure. The 10 W target follows A = P_target/P'' with a twelve-plate CAD cap; default L_p under-predicts at 1.66 W. Bench T1 (±30%) must validate L_p or revise P''. Mode A acoustic harvest remains mW-class; Mode B net gain requires flux gain g to exceed ultrasonic parasitics. The 22.8 MW/km² column median is exploratory and must not be conflated with bench claims.

6. Conclusion

We presented a physics-first CHORUS framework and a concrete PRO bench skid (SGH-1) for anthropogenic brine gradients. Near-term work centers on PRO, DAQ, and CP/ultrasonic assist; column-scale and TSC claims are tiered as context. Every equation is reproducible from differential-harness open-source artifacts.

Appendix A — Governing equations (Layer F PRO)

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π = iRTc

Δπ = iRT(c_draw − c_feed)

V■_w = L_p A(Δπ − ΔP)

ΔP* = Δπ/2

P_elec = η_mem η_hyd ρ V■_w ΔP

c_w/c_b = exp(J_w/k_m)
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Appendix B — Repository artifacts

Path	Role
docs/CHORUS_MATH_PLAN.md	Master equation list
docs/math/PRO_LAYER_DERIVATION.md	Layer F derivation
notebooks/CHORUS_physics_proof.ipynb	Executable proof
exports/chorus_results.json	Column + estuary numbers
exports/sgh1_design.json	Skid sizing
hardware/openscad/	23-part CAD library

References

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[4] Fang et al., Energy Environ. Sci. (2026) — PV–MHD coupling.

[5] Yao et al., Advanced Materials — moisture charge separation.

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